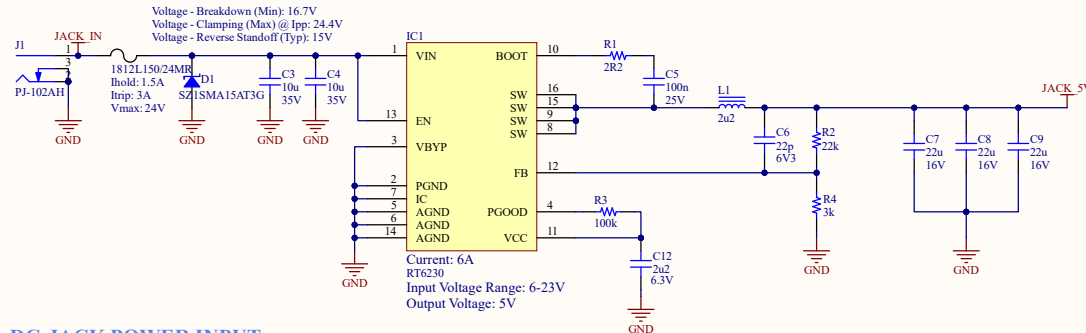


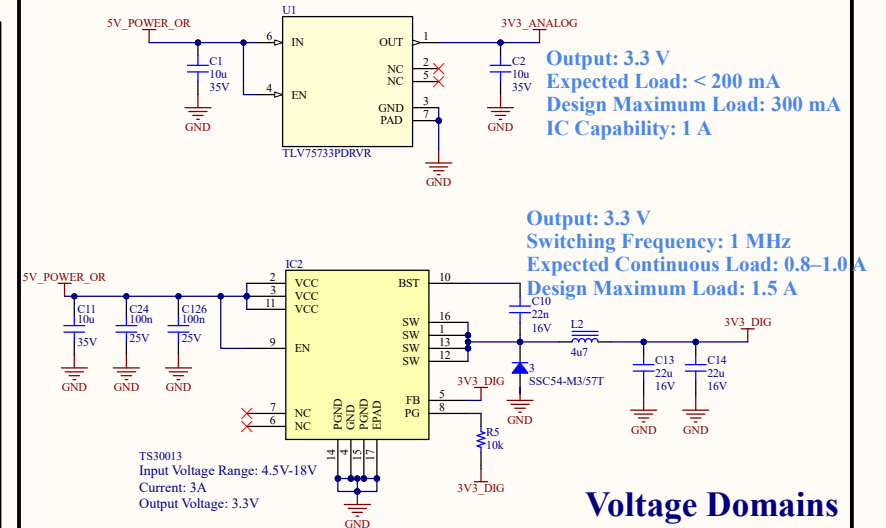
Power Section



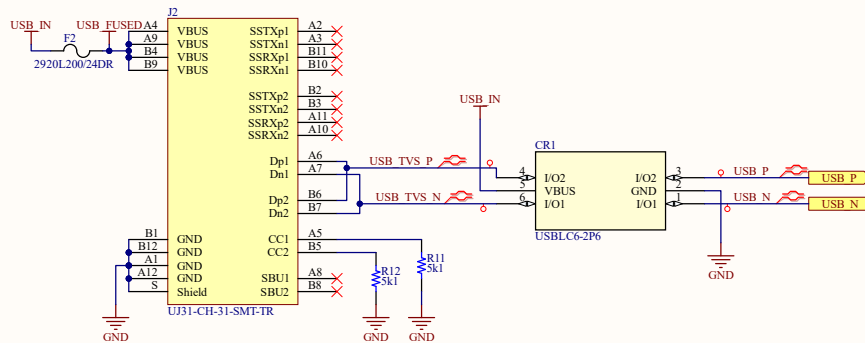
DC-JACK POWER INPUT
9-12 VDC REGULATED, CENTER-POSITIVE
Recommended Adapter: 9 V / 2 A or 12 V / 1.5 A
JACK_5V: 5.0 V / 1.5 A CONTINUOUS
2.0 A PEAK

**USE CENTER-POSITIVE
ADAPTER ONLY, NO
REVERSE-POLARITY
PROTECTION !!!**

DC-Jack Input

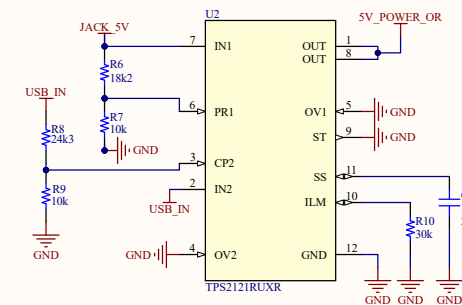


Voltage Domains



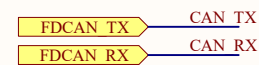
**USE A 5 V TYPE-C POWER SOURCE ONLY.
NO USB POWER DELIVERY SUPPORT.
DO NOT TRY HIGH-CURRENT OPERATION
FROM AN UNKNOWN PC OR USB CHARGER.**

USB-TYPE-C Input



Power MUX

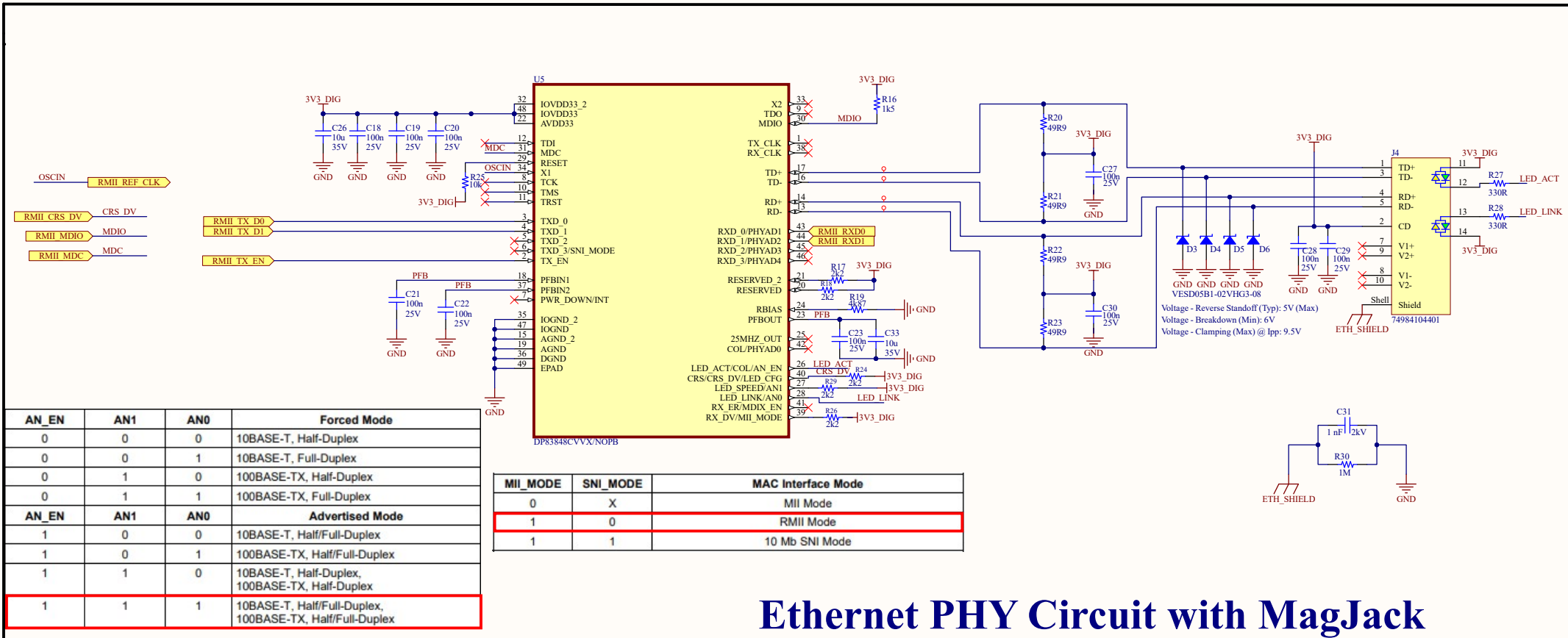
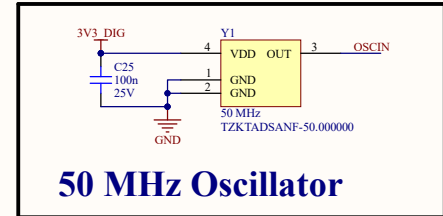
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A3		
Date:	8.16.2026	Sheet of
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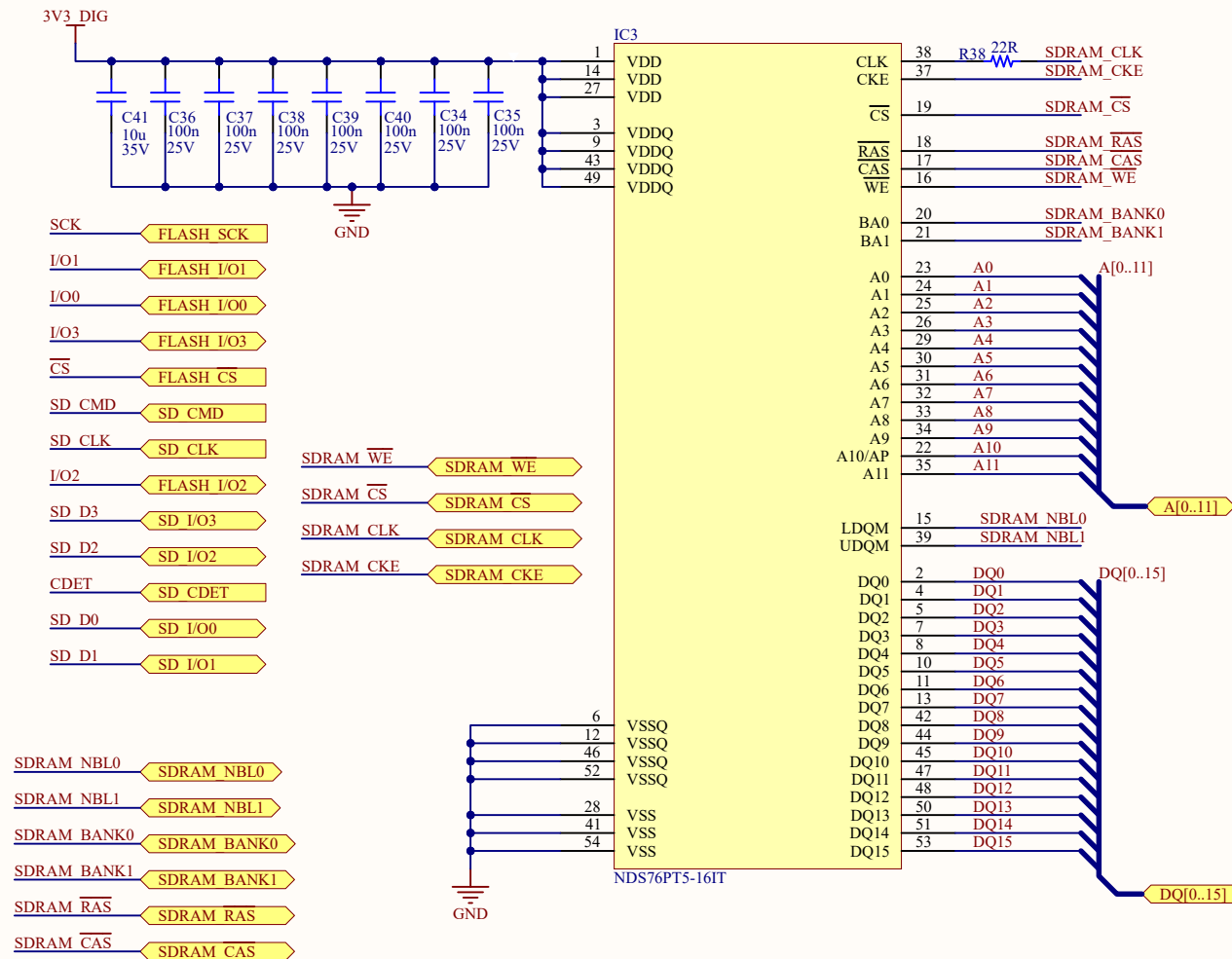
Ethernet Section



Ethernet PHY Circuit with MagJack

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File:	C:\Users\...\Ethernet.SchDoc	Drawn By:

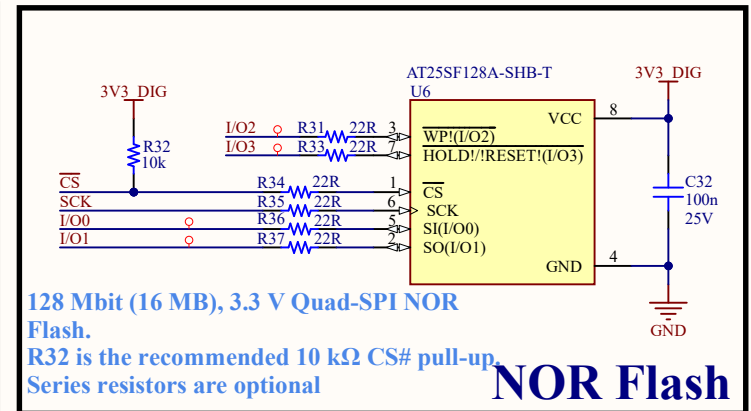
Memory Section



128 Mbit (16 MB) SDR SDRAM, x16 data bus, 4 internal banks.

R38 can be configured according to the signal integrity of SDRAM_CLK signal.

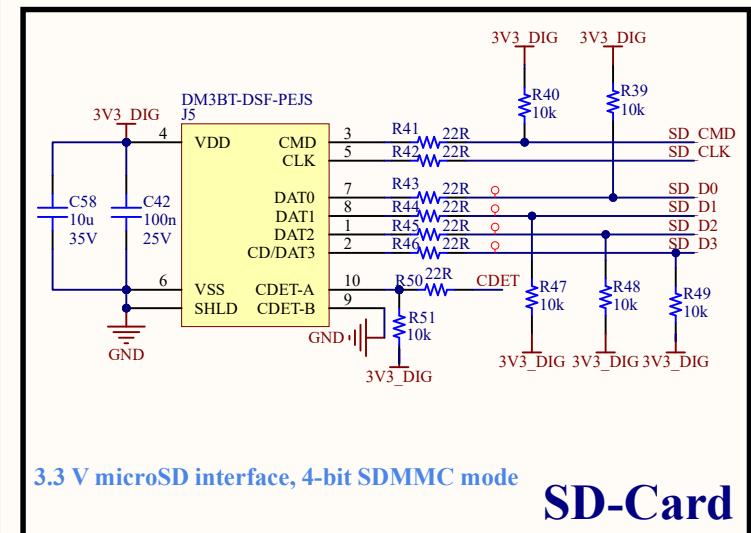
SDRAM



128 Mbit (16 MB), 3.3 V Quad-SPI NOR Flash.

R32 is the recommended 10 kΩ CS# pull-up. Series resistors are optional

NOR Flash



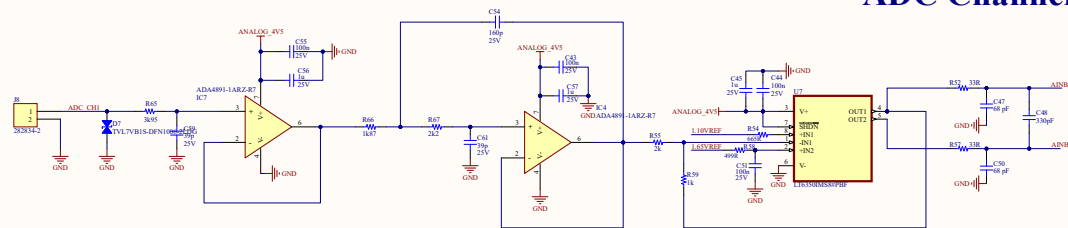
3.3 V microSD interface, 4-bit SDMMC mode

SD-Card

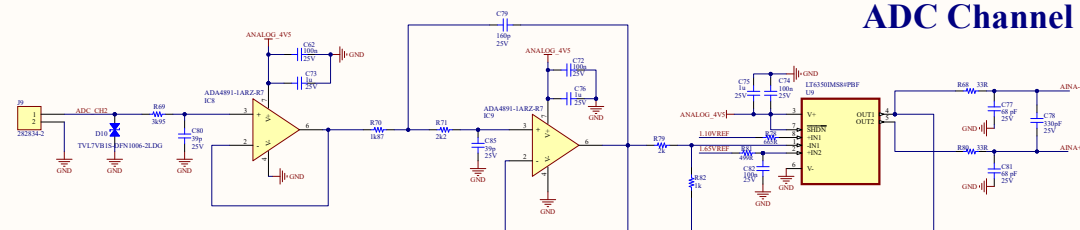
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Analog Circuitry

ADC Channel 1



ADC Channel 2

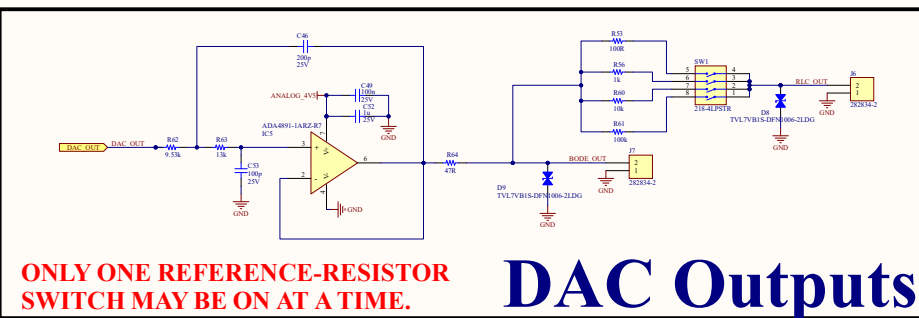


16-bit simultaneous sampling at up to 4 MSPS per channel. External input range: 0–3.3 V, single-ended. Input RC cutoff ≈ 1.03 MHz; second-order active low-pass cutoff ≈ 125 kHz. Intended precision measurement band: 10 Hz–50 kHz. ADC input network: 33 Ω / 68 pF / 330 pF.

Do not apply negative voltage or signals above 3.3 V. **ADC Inputs**

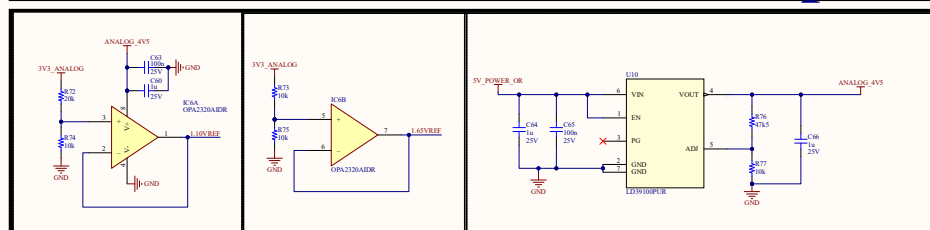
Two simultaneously sampled, 16-bit channels operate at up to 4 MSPS. Each external single-ended input is protected, buffered, low-pass filtered, level-shifted, and converted to a differential signal centered at 1.65 V before driving the AD7380. The ADC input networks use the manufacturer-recommended 33 Ω / 68 pF / 330 pF differential RC configuration. The nominal external input range is 0 V to 3.3 V referenced to board GND.

The integrated DAC signal path provides programmable excitation for frequency-response and impedance measurements. BODE_OUT is the filtered excitation output; RLC_OUT is the node after the manually selected reference resistor. Select only one reference resistor at a time. External test equipment and the device under test must share board GND. The analog interfaces are not isolated and are not permanently 50 Ω terminated.



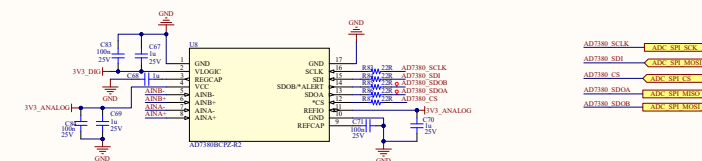
ONLY ONE REFERENCE-RESISTOR SWITCH MAY BE ON AT A TIME.

DAC Outputs



The adjustable LD39100 generates a nominal 4.5 V analog rail from 5V_POWER_OR. Buffered 1.10 V and 1.65 V bias references are derived from 3V3_ANALOG using an OPA2320. The 1.65 V node defines the differential-driver output common-mode voltage, while the 1.10 V node provides the input level-shift reference.

Voltage References



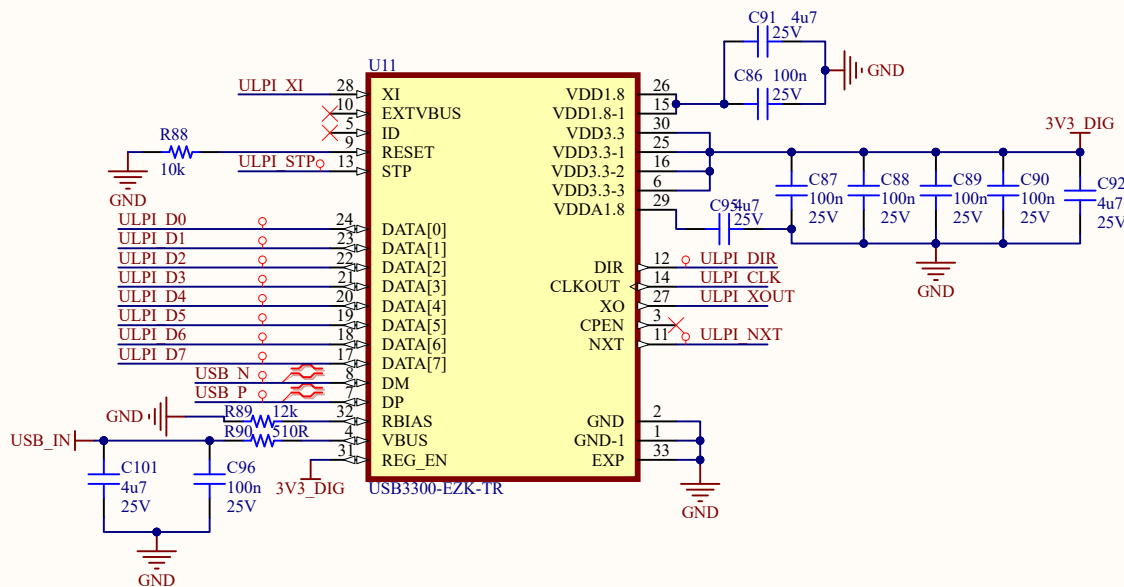
Dual simultaneous-sampling, 16-bit SAR ADC operating at up to 4 MSPS per channel. VCC is supplied from 3V3_ANALOG and VLOGIC from 3V3_DIG. REFIO is driven by the external 3.3 V analog reference rail; firmware must select external-reference operation before precision conversions.

ADC

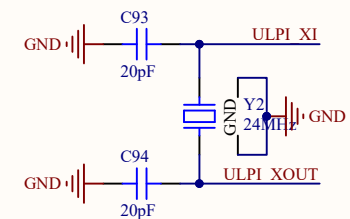
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USB-HS PHY Section

The USB3300 provides the external USB 2.0 High-Speed physical layer for the STM32H735 through the 8-bit ULPI interface. The PHY operates from the 3.3 V digital supply and uses its internal regulators to generate separate 1.8 V digital and analog rails. A 24 MHz crystal provides the PHY reference clock, while the USB3300 supplies the 60 MHz ULPI clock to the MCU. The interface is configured for USB device-only operation. OTG ID detection, host VBUS drive control, and external VBUS indication are not used. USB DP/DM pull-up, pull-down, and High-Speed termination resistors are integrated inside the PHY.

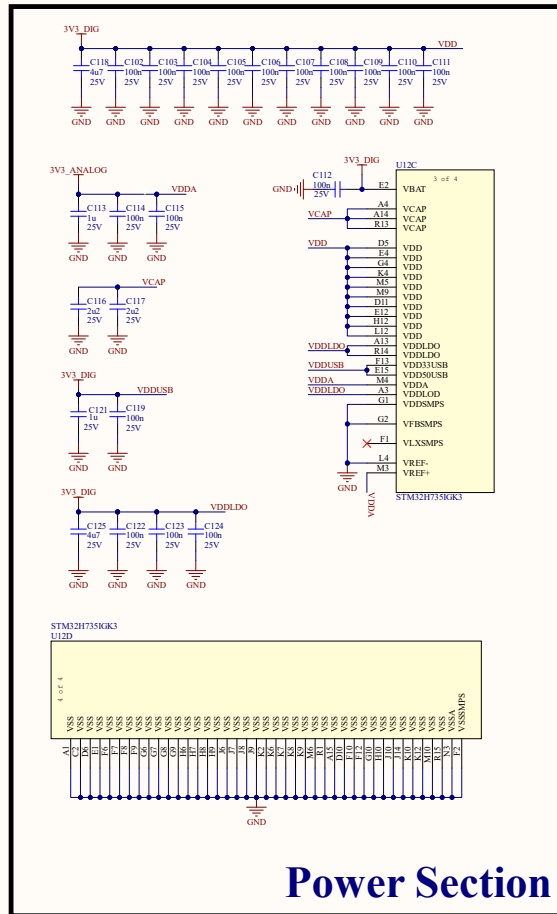
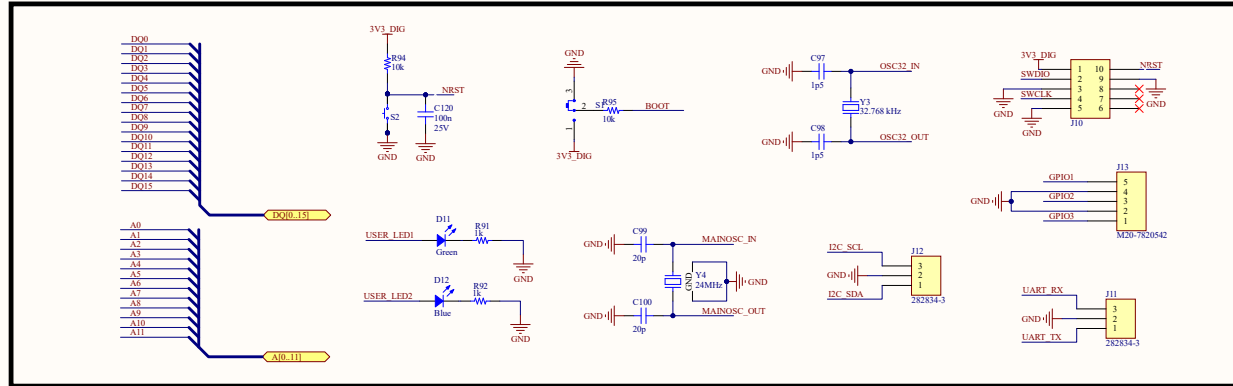


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USB N	USB N
ULPI D0	ULPI D0
ULPI D1	ULPI D1
ULPI D2	ULPI D2
ULPI D3	ULPI D3
ULPI D4	ULPI D4
ULPI D5	ULPI D5
ULPI D6	ULPI D6
ULPI D7	ULPI D7
ULPI STP	ULPI STP
ULPI DIR	ULPI DIR
ULPI NXT	ULPI NXT
ULPI CLK	ULPI CLK

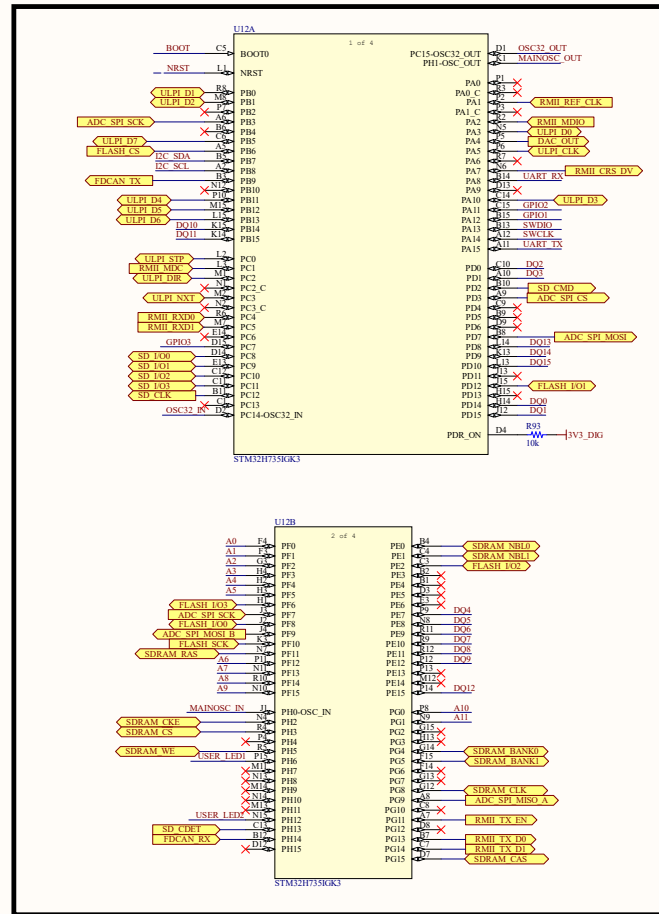


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Microcontroller Section



Power Section



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